Interference

> **SYNOPSIS**
>
> Send light through two narrow slits and the screen does not show two bright patches. It shows a row of bright and dark bands — including **dark bands where light from both slits arrives**. No amount of ray tracing accounts for that. This note is the geometry that does, and it comes down to one length: the difference in how far the two waves have travelled.

BEFORE YOU START — PREDICT

Light arrives at a point on the screen from **both** slits. Twice the light. Under what circumstance could that point be **completely dark**? Answer in one sentence before you read on.

PERFORMANCE EXPECTATIONS

1. **Distinguish** constructive from destructive interference by the phase relationship and by the path difference.
2. **Derive** $\Delta l = d \sin\theta$ from the geometry of two distant parallel rays.
3. **Apply** $d \sin\theta = m\lambda$ and $d \sin\theta = (m + \tfrac{1}{2})\lambda$ to locate bright and dark fringes.
4. **Use** the small-angle result $y_m = m\lambda D/d$, and say when it is allowed.

1. Interference

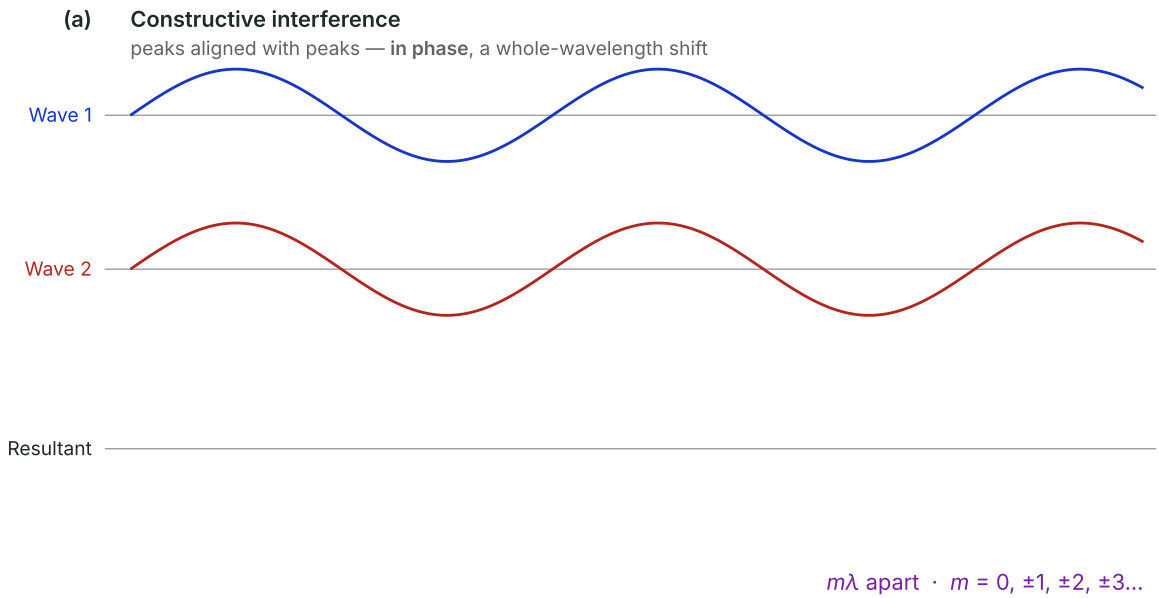


Figure 1. Two waves arriving **in phase**. The resultant is drawn from the sum of the two above it, so its amplitude really is 2*X* — nothing about the picture was adjusted to make that come out.

Your job on Figures 1 and 2. Wave 1 and Wave 2 are drawn for you in both panels. Add the resultant on the bottom axis of each — add the two waves point by point rather than guessing the shape. On Figure 3, mark the extra distance the bottom ray travels and write down what it equals.

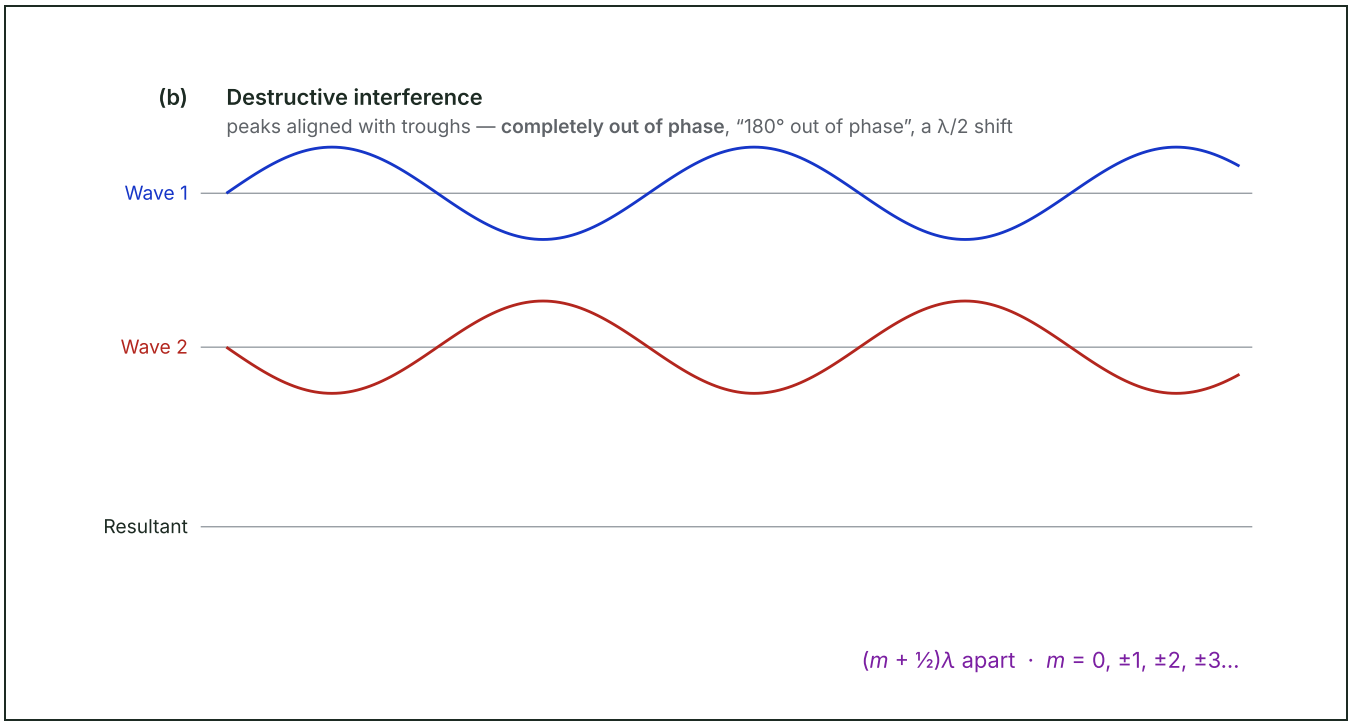


Figure 2. The same two waves, one shifted by half a wavelength. The resultant is again the sum, and it is flat: the light has gone. **That is the thing a ray picture cannot do**, and it is why this note exists.

So the condition is a statement about **how far apart the two waves are** when they meet:

- **Constructive** — the waves are $m\lambda$ apart, $m = 0, \pm1, \pm2, \pm3\ldots$
- **Destructive** — the waves are $(m + \tfrac{1}{2})\lambda$ apart, $m = 0, \pm1, \pm2, \pm3\ldots$

TRY IT YOURSELF

The PhET wave-interference simulation lets you set the two sources going and watch the pattern build: phet.colorado.edu/en/simulations/wave-interference

Two slits

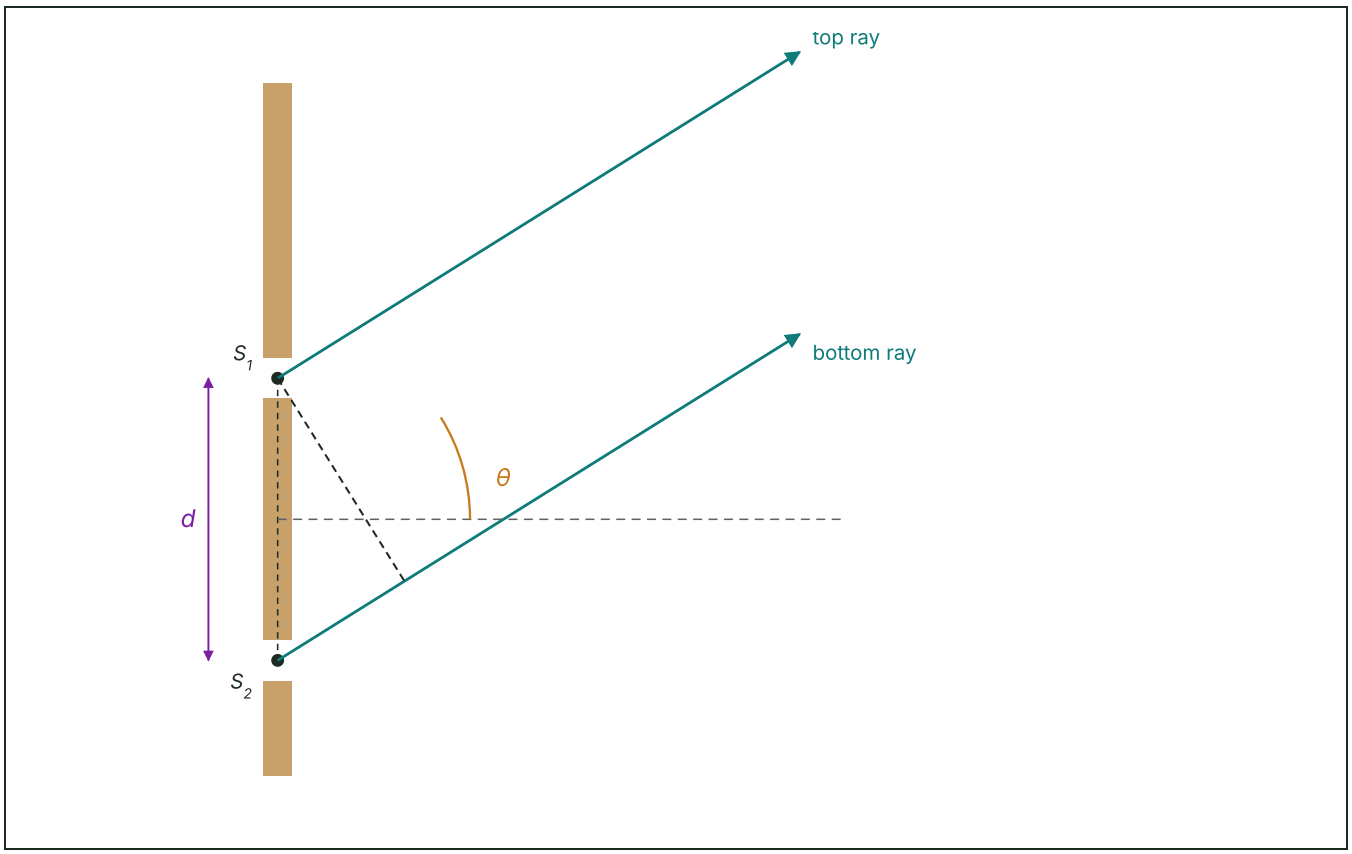


Figure 3. Why $\Delta l = d \sin\theta$. The screen is so far away compared with d that the two rays heading for the same point are effectively **parallel**. Drop a perpendicular from S_1 onto the other ray and the bit left over — drawn in red — is the extra distance the bottom ray travels. It is the opposite side of a right triangle with hypotenuse d, so it is $d \sin\theta$ exactly.

Δl is the **difference in path lengths between the top and bottom rays**. Feed it into the two conditions above and the whole of two-slit interference falls out:

THE TWO CONDITIONS

$$d \sin\theta = m\lambda \quad \text{(constructive)}$$

$$d \sin\theta = (m + \tfrac{1}{2})\lambda \quad \text{(destructive)}$$

$m = 0, \pm1, \pm2, \pm3\ldots$

***m* is the order of the interference.** $m = 0$ is the zeroth order — the central bright fringe; $m = \pm1$ is first order, $m = \pm2$ second order, and so on outward in both directions.

Where the fringe lands on the screen

A bright fringe is at $d \sin\theta = m\lambda$. **For small angles** $\sin\theta \approx \theta \approx \tan\theta$, and $\tan\theta = y/D$, so

$$d \sin\theta = m\lambda \;\rightarrow\; d\,(y_m/D) = m\lambda\, y_m = m\lambda D \,/\, d$$

which says the bright fringes are **evenly spaced** on the screen — as long as the angle stays small enough for the approximation to hold.

Worked example

A **HeNe laser** shines on a double slit with d = **0.01 mm**. The **third bright fringe** appears at θ_3 = **10.95°**. What is the wavelength of the laser?

1. Work it before opening the solution. Two decisions settle it: which of the two conditions applies, and what m is.

THE PUNCHLINE

$\Delta l = d \sin\theta$, and everything follows from it

Back to the prediction: the point is dark when the two waves arrive **half a wavelength out of step** — light plus light giving no light. A ray cannot be half a wavelength out of step with another ray, because a ray has no wavelength. This is the first result in the module that **requires** light to be a wave.

Looking ahead

Two slits give a row of evenly spaced fringes. **Note 11** closes *one* of the slits — and finds that a single slit still produces a pattern of bright and dark bands, for a reason that turns the rule of this note upside down.

2. Light of 600 nm falls on slits 0.05 mm apart. At what angle, in degrees, is the **first** bright fringe?

[ ]°

3. In $y_m = m\lambda D/d$, what has been assumed?

◯ That the light is monochromatic. ◯ That the angle is small, so sin θ ≈ tan θ. ◯ That the two slits are of equal width. ◯ That the screen is closer than the slit separation.

4. Slits 0.10 mm apart, a screen 2.0 m away, light of 500 nm. How far from the centre, in mm, is the **second** bright fringe?

mm

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.